



Extending thinking about sequences

1 An arithmetic (or linear) sequence is a sequence where the difference between successive terms is a _____. Tick **1** correct answer

- ☐ constant
- ☐ multiplier
- ☐ negative number
- ☐ positive number
- ☐ variable

2 Find the next term in the arithmetic sequence -25, -18, -11, -4, _____.
Fill in the blank

3 Find the missing term in the arithmetic sequence 27, _____, 55, 69, 83,
... Fill in the blank

4 Find the common ratio of the geometric sequence 0.4, 1.6, 6.4, 25.6, ... Tick **1** correct answer

- ☐ 0.25
- ☐ 0.4
- ☐ 0.6
- ☐ 4
- ☐ 6

5 Find the next term in the geometric sequence 0.4, 1.6, 6.4, 25.6, _____,
... Give your answer as a decimal. Fill in the blank



6 Which statements are true of the sequence 1920, 960, 480, 240, 120, ... ? Tick 2 correct answers

- ☐ It is a decreasing sequence therefore will contain negative values in the future
- ☐ It is a geometric sequence with a common ratio greater than 1
- ☐ It is a geometric sequence with a common ratio less than 1
- ☐ It will eventually start to increase again.
- ☐ It will never reach zero.

